

No. of Functional Features Included in the Solution

Date	4 October 2026
Team ID	6ab2951f1266c775575a6b80
Project Name	Farm Management System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution in the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Authentication	Secure user login and role management (e.g., Farmer, Manager, Analyst).	Auth	Done	0.5
2	Field Asset Logging	Create and manage detailed profiles for farm fields, including GIS boundaries.	Field Mgmt	Done	1.0
3	Equipment Inventory	Track machinery, tools, and maintenance schedules.	Inventory	Done	0.75
4	Task Scheduling	Assign farm tasks (e.g., planting, watering) with timelines.	Operations	Done	0.75
5	Sensors Integration	Connect and read data from real-time soil and weather sensors.	IoT Integration	Done	0.75
6	Crop Planning and Cycle Tracking	Plan crop rotations and track growth stages.	Agronomy	Done	0.5
7	Weather Data API	Fetch and display historical and predictive weather data.	External Services	Done	0.25
8	Harvest Reporting Dashboard	Generate reports on yield and production for all fields.	Reporting	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	10
Total Features Implemented	8
Core / Must-Have Features	5 (e.g., Auth, Field Mgmt, Task, Sensors, Harvest)
Additional / Nice-to-Have Features	3 (e.g., Inventory, Crop Plan, Weather API)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	8	e.g., login screen, map view, equipment forms, dashboard views
2	Backend / Logic	8	e.g., auth service, sensor data processor, task assignment logic, report generators
3	Database / Storage	8	e.g., user DB, field DB with PostGIS, equipment DB, sensor data time-series DB
4	API / Integration	2	e.g., IoT sensor gateway API, AccuWeather API
5	Security / Authentication	1	e.g., JWT-based auth and RBAC